

Q1. JEE Main 2026 (23 January Shift 1)

In a screw gauge, the zero of the circular scale lies 3 divisions above the horizontal pitch line when their metallic studs are brought in contact. Using this instrument thickness of a sheet is measured. If pitch scale reading is 1 mm and the circular scale reading is 51 then the correct thickness of the sheet is _____ mm.

[Assume least count is 0.01 mm]

- (1) 1.50 (2) 1.48 (3) 1.54 (4) 1.51

Q2. JEE Main 2026 (28 January Shift 2)

In an experiment, a set of reading are obtained as follows - 1.24 mm, 1.25 mm, 1.23 mm, 1.21 mm. The expected least count of the instrument used in recording these readings is _____ mm.

- (1) 0.001 (2) 0.1 (3) 0.01 (4) 0.05

Q3. JEE Main 2026 (24 January Shift 2)

In a vernier callipers, 50 vernier scale divisions are equal to 48 main scale divisions. If one main scale division = 0.05 mm, then the least count of the vernier callipers is _____ mm.

- (1) 0.005 (2) 0.05 (3) 0.002 (4) 0.02

Q4. JEE Main 2026 (28 January Shift 1)

When both jaws of vernier callipers touch each other, zero mark of the vernier scale is right to zero mark of main scale, 4th mark on vernier scale coincides with certain mark on the main scale. While measuring the length of a cylinder, observer observes 15 divisions on main scale and 5th division of vernier scale coincides with a main scale division. Measured length of cylinder is _____ mm. (Least count of Vernier calliper = 0.1 mm)

- (1) 15.5 (2) 15.9 (3) 15.1 (4) 15.4

ANSWERS AND SOLUTIONS

1. (3)

2. (3)

3. (3)

4. (3)

1. (3) For a screw gauge, the measurement formula is: Reading = Pitch scale + (Circular scale $\times$ Least count) - Zero error

The zero of circular scale lies 3 divisions above the pitch line, indicating negative zero error:

$$\text{Zero error} = -3 \times 0.01 = -0.03 \text{ mm}$$

$$\text{Reading} = 1 + (51 \times 0.01) - (-0.03) = 1 + 0.51 + 0.03 = 1.54 \text{ mm}$$

2. (3) Least count is the smallest division readable on an instrument scale.

Given readings are 1.24 mm, 1.25 mm, 1.23 mm, 1.21 mm, all expressed to two decimal places.

The precision of measurement shows differences as small as 0.01 mm (e.g., $1.25 - 1.24 = 0.01 \text{ mm}$).

This indicates all readings can be recorded to the nearest 0.01 mm, which represents the smallest measurable division.

Therefore, the least count of the instrument is 0.01 mm.

3. (3) Given: 50 vernier divisions = 48 main divisions and 1 main division = 0.05 mm

$$50 \text{ vernier divisions} = 48 \times 0.05 = 2.4 \text{ mm}$$

$$1 \text{ vernier division} = 2.4/50 = 0.048 \text{ mm}$$

$$\text{Least count} = 1 \text{ main division} - 1 \text{ vernier division} = 0.05 - 0.048 = 0.002 \text{ mm}$$

4. (3) Given Least Count (LC) = 0.1 mm.

When jaws touch, the zero of the vernier scale is to the right of the main scale zero, indicating a positive zero error.

Zero error (e) = $+(n \times LC)$, where n is the coinciding vernier division.

$$e = +(4 \times 0.1) = +0.4 \text{ mm.}$$

While measuring the cylinder:

Main Scale Reading (MSR) = 15 divisions.

Assuming 1 MSD = 1 mm (standard for $LC = 0.1 \text{ mm}$), $MSR = 15 \text{ mm}$.

Vernier Scale Reading (VSR) = $5 \times LC = 5 \times 0.1 = 0.5 \text{ mm}$.

Observed Reading = $MSR + VSR = 15 + 0.5 = 15.5 \text{ mm}$.

True Measured Length = Observed Reading - Zero Error

$$\text{True Length} = 15.5 - 0.4 = 15.1 \text{ mm.}$$